

Financial Performance & Discount Sensitivity Dashboard – Power BI Case Study

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1 Executive Summary

This case study documents the design and delivery of a two-page, executive-grade Power BI dashboard – **Executive Overview** and **Discount Sensitivity Analysis** – built to diagnose and reverse margin erosion caused by uncontrolled promotional discounting. Working with a star-schema model covering Orders, Products, Customers, Marketing, and Calendar dimensions over the period March 2025 – February 2026, the project delivered a P&L Waterfall analysis from gross revenue to net profit, discount-sensitivity scatter analysis by product category and marketing channel, and a family of DAX measures (Effective Discount%, DiscountOrders%, Discount ROI, Adjusted_Discount) that quantify how discount spend converts – or fails to convert – into revenue and profit. The headline picture: **USD 2.81B total revenue at a 16.44% net profit margin, with 56.69% of all orders sold under discount** – a clear margin-leakage problem and a concrete opportunity to shift from blanket discounting to a tiered, elasticity-based discount policy.

2 Business Context & Scope

The organization faced a classic revenue–profit paradox: top-line revenue was growing, yet net profit margin was being eroded because discounting had become a default sales lever rather than a targeted instrument. With more than half of all orders (56.69%) transacted under discount, leadership had no reliable way to answer three critical questions: (1) how much revenue is being given away, (2) which product categories and marketing channels actually pay back that discount spend, and (3) what happens to net profit margin if discount policy changes. The existing reporting showed *what* was sold, but not *why* profit was leaking. The objective of this project was therefore to build a decision-support dashboard that makes discount economics transparent and testable – turning a static sales report into a sensitivity-analysis tool for pricing and marketing decisions. The scope covered two report

pages built on a standardized 1600×900 (16:9) canvas: an **Executive Overview** for leadership and a **Discount Sensitivity** page for pricing and marketing teams.

3 Technical Approach & Implementation

Data modeling. The semantic model follows a star-schema architecture: a fact table of orders surrounded by dimension tables – Products, Customers, Marketing (channel), Calendar, and a P&L Structure table that sequences the profit-and-loss walk. Fact and dimension tables are separated with one-to-many relationships, keeping DAX filter context predictable across both pages. Data-quality hygiene was enforced: tables were renamed consistently (e.g., Calendar), KPI cards were de-duplicated, and the ordinal Discount Band column received a sort-by-column order so discount ranges rank correctly on axes.

DAX measure engineering. A dedicated measure table (Measure_List) hosts the analytical layer, built with best-practice patterns (VAR/RETURN, DIVIDE, SUMX iterators):

Measure	Purpose
Total Revenue, NetProfit, NetProfitMargin%	Core profitability KPIs
Adjusted_Discount (Total Discount)	Total discount dollars granted
Effective Discount%	Discount spend as a share of revenue
DiscountOrders%	Share of orders sold under discount
Discount ROI	Return of discount spend against profit contribution
P&L Waterfall Value	Sequenced P&L walk from gross revenue to net profit

Report architecture. *Executive Overview* leads with KPI cards, a **Waterfall chart** decomposing revenue into net profit, a **combo chart** of Total Revenue (columns) vs. Net Profit Margin % (line) across discount bands, and a channel slicer. *Discount Sensitivity* adds two **scatter plots** (Adjusted_Discount vs. NetProfitMargin%, sized by revenue, colored by category and channel), bar charts of discount penetration by channel and city, and Discount ROI cards. Every visual carry an action-oriented title (e.g., “Discount, NetProfit and %DiscountOrders by City”) stating the question it answers.

4 Key Analytical Findings

1. Margin leakage concentrates in high discount bands. The P&L Waterfall shows that the largest single erosion between gross revenue and net profit is discounting – not logistics or operating overhead. In discount bands above ~20%, incremental order volume does not

compensate for the price concession: the revenue column keeps growing while the margin line on the same combo chart bends downward.

2. Discount sensitivity differs sharply by channel. The channel scatter reveals that customers arriving through organic/direct channels convert at healthy margins even with minimal discounting — discounts there largely burn margin without adding conversion. Paid and affiliate-driven channels, by contrast, show stronger discount dependence, but their final margins sit near break-even once discount spend is fully attributed.

3. Product categories respond heterogeneously. The category scatter (sized by revenue) separates fast-turning, high-AOV categories where discounting acts as a profitable volume lever from thin-margin categories where any discount increase translates almost one-to-one into net-profit decline — the primary source of the 16.44% net margin under 56.69% discount penetration.

4. Discount ROI is not uniform. The Discount ROI card makes it explicit that a dollar of discount does not buy the same amount of profit everywhere; pooling all discounts into one blended rate hides where spend is productive and where it is leakage.

5 Recommendations

Each recommendation below is grounded in the sensitivity evidence from the dashboard:

1. **Replace blanket discounts with tiered discounting.** Gate discounts on basket value (e.g., a threshold above the current AOV) to lift average order value and stop giving margin away on orders that would have converted anyway — especially in organic/direct channels.
2. **Set a discount cap policy by category.** For thin-margin categories, cap discounts at the band where the margin curve inverts (near ~12% in this dataset) and route any further promotion through bundle or volume mechanics instead of price cuts.
3. **Reallocate marketing spend on a margin basis.** Eliminate public coupon codes in high-converting organic channels; reserve discount codes for re-engagement and churn-prevention campaigns where elasticity is proven.
4. **Monitor Margin Gap weekly.** Track Effective Discount% against a target rate and alert when it drifts above target, so discount-driven margin surprises surface mid-month rather than a month-end close.
5. **Adopt the Discount Action Matrix.** Classify each category × channel cell by discount elasticity and net margin into four quadrants — Growth Stars, Profit Engines, Volume Drivers, and Profit Traps — and assign a distinct discount posture to each.

6 Project: Financial Performance & Discount Sensitivity Dashboard (Power BI)

- **Situation:** Revenue was growing while net profit margin eroded — 56.69% of orders were sold under discount with no tooling to test whether discount spend paid back.
- **Task:** Build an executive decision-support dashboard that quantifies discount economics and simulates the margin impact of discount policy changes.
- **Action:** Designed a star-schema data model (fact/dimension separation, sort-by-column ordering); engineered a DAX measure layer (Effective Discount%, DiscountOrders%, Discount ROI, P&L Waterfall Value) using VAR/RETURN and DIVIDE best practices; built a P&L Waterfall for revenue-to-profit decomposition and scatter-based discount-elasticity analysis by product category and marketing channel on a standardized 16:9 report canvas.
- **Result:** Revealed margin leakage concentrated in discount bands above ~20% and near break-even margins on paid channels; surfaced a measurable margin-recovery opportunity by capping thin-margin discounts and shifting to tiered discounting.
- **Impact:** Converted a static sales report into a discount decision-support system — giving pricing and marketing teams an elasticity-based framework (Discount Action Matrix) for protecting net profit without sacrificing volume.

Skills to list alongside this project: Power BI · DAX (Measures, CALCULATE, SUMX, Time Intelligence) · Data Modeling (Star Schema, Fact & Dimension Tables) · Data Visualization (Waterfall, Scatter, KPI Cards, Slicers) · Dashboard Design · Financial & P&L Analysis · Discount/Pricing Analytics.